**Title:** ***Experiment No. 4 – Frequency Domain Filtering***

**Subject:** Computer Vision Lab

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**Aim:** *To perform frequency domain filtering using Fourier Transform (DFT, LPF, HPF).*

**CODE**

```
import cv2

import numpy as np

import matplotlib.pyplot as plt

import os
```

**# Step 1: Load grayscale image**

```
file_path = "/workspaces/computer-vision-Experiments./download.jpg"   # apna image path yahan daalo

img = cv2.imread(file_path, 0)

if img is None:

   raise ValueError("Image not loaded. Check file path.")

rows, cols = img.shape
```



**# Step 2: Compute DFT**

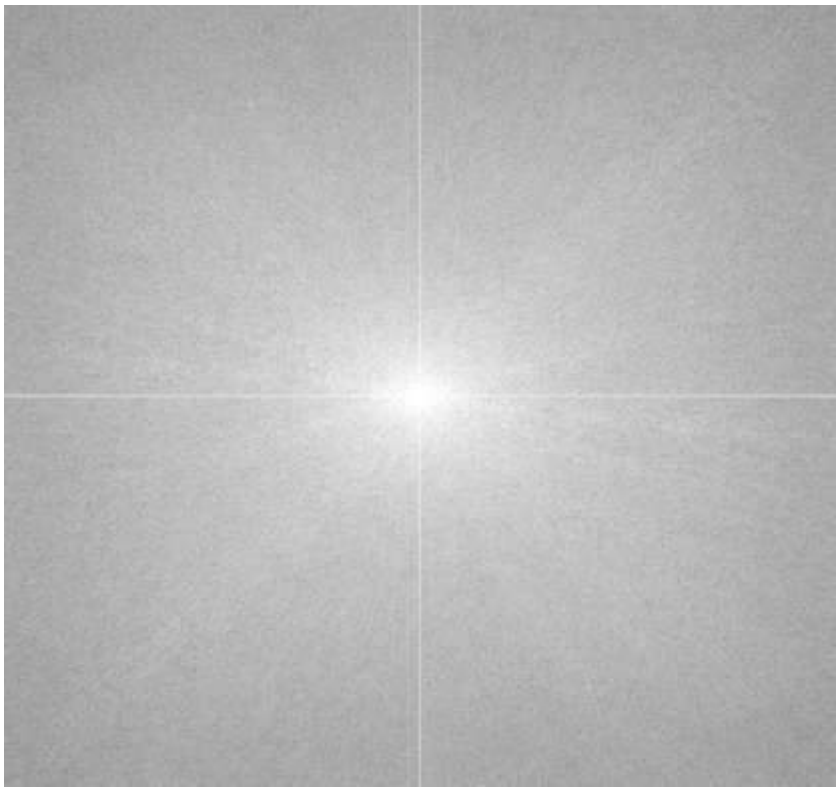
```
dft = cv2.dft(np.float32(img), flags=cv2.DFT_COMPLEX_OUTPUT)
```

**# Step 3: Shift zero-frequency to center**

```
dft_shift = np.fft.fftshift(dft)
```

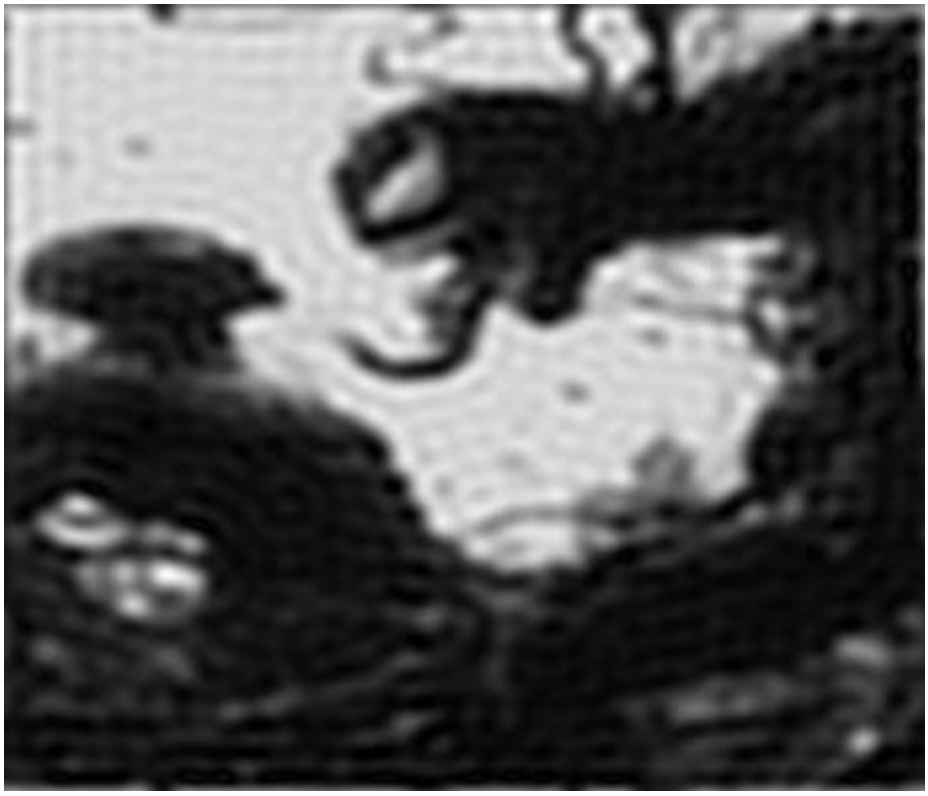
**# Step 4: Magnitude Spectrum**

```
magnitude_spectrum = 20*np.log(cv2.magnitude(dft_shift[:,:,0], dft_shift[:,:,1]))
```



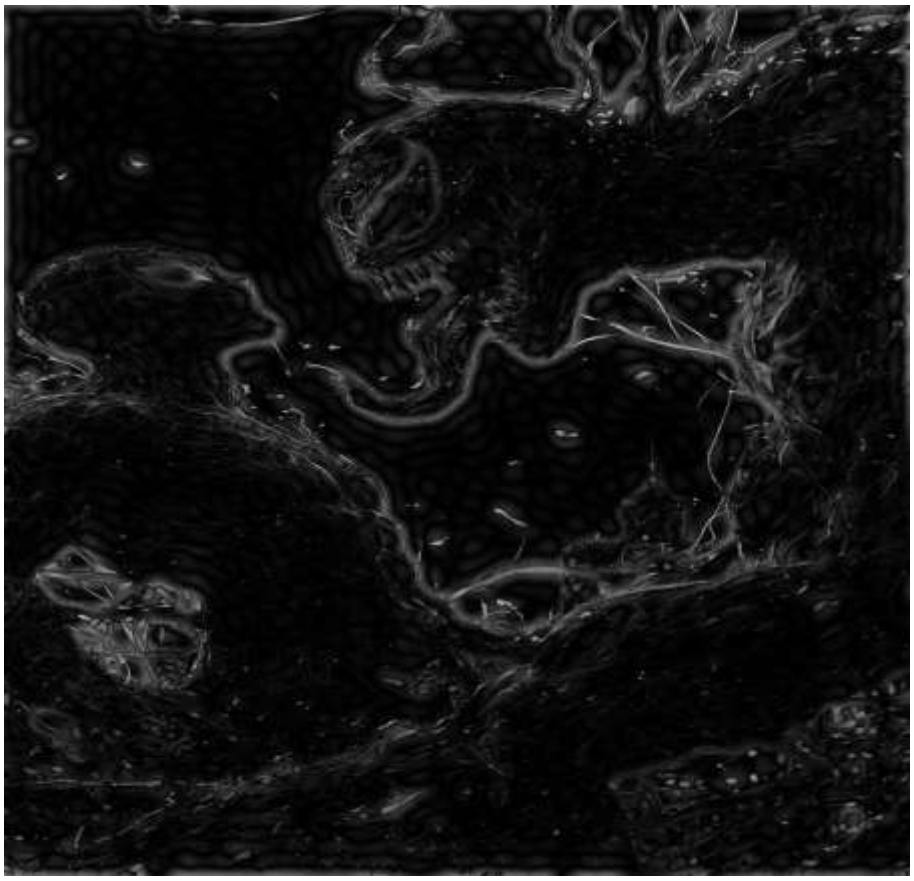
**# Step 5: Low-Pass Filter**

```
mask = np.zeros((rows, cols, 2), np.uint8)
crow, ccol = rows//2 , cols//2
r = 30  # radius of LPF
cv2.circle(mask, (ccol, crow), r, (1,1), -1)
lpf = dft_shift * mask
```



**# Step 6: High-Pass Filter**

hpf = dft_shift * (1 - mask)



```
# Step 7: Inverse DFT function
def idft_reconstruct(filtered):
    f_ishift = np.fft.ifftshift(filtered)
    img_back = cv2.idft(f_ishift)
    img_back = cv2.magnitude(img_back[:,:,0], img_back[:,:,1])
    return img_back
img_lpf = idft_reconstruct(lpf)
img_hpf = idft_reconstruct(hpf)

# Step 8: Save outputs separately
os.makedirs("outputs", exist_ok=True)
cv2.imwrite("outputs/original.png", img)
cv2.imwrite("outputs/magnitude_spectrum.png", magnitude_spectrum)
cv2.imwrite("outputs/lpf.png", img_lpf)
cv2.imwrite("outputs/hpf.png", img_hpf)
print(" ✅ Separate image files saved in outputs/ folder:")
print(" - original.png")
print(" - magnitude_spectrum.png")
print(" - lpf.png")
print(" - hpf.png")

# Step 9: Analyze impact
print("\n--- Analysis ---")
print("Low-Pass Filter: Image becomes smoother, noise reduced, but edges blurred.")
print("High-Pass Filter: Edges and fine details enhanced, but noise may increase.")
```

**# Step 10: Document observations**

observations = """

Observations:

1. Fourier Transform converts image into frequency domain for selective filtering.

2. Low-Pass filtering suppresses high-frequency noise, improves smoothness.

3. High-Pass filtering enhances edges and fine details, useful for feature extraction.

4. Frequency domain filtering is efficient for large kernel operations compared to spatial domain.

5. Practical applications include medical imaging, satellite image enhancement, and biometric systems.

"""print(observations)